

Summary

Dynamic Data Scientist with 3.5 years of experience in SQL, deep learning algorithms, cloud services, and automated system monitoring. Proven track record in enhancing operational efficiency, developing predictive models, and analyzing complex datasets. Adept at collaborating with cross-functional teams and leveraging advanced statistical methods to drive data-driven decision-making. Proficient in Python, SQL, Excel and various machine learning frameworks, and cloud platforms. Strong problem-solving and analytical skills and a commitment to innovation and excellence in delivering actionable insights.

Skills & Abilities

Programming Languages/Web Technologies: Python, SQL, R, MATLAB, SAS, JavaScript.
Databases & Cloud Services: MySQL, NoSQL, REDCap, **Google Cloud Platform (GCP)**, Amazon Web Services(AWS).
Libraries & Frameworks: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, OpenCV, TensorFlow, Keras, PyTorch, PyQt5.
Data Science & Analytics Tools: **Visual Studio Code**, Microsoft Excel, **Tableau**, GitHub, Machine Learning (ML), Natural Language Processing (NLP), Large Language Models(LLMs), Statistical Analysis, Stata, Data Visualization, Data Management, **ETL** .
Certifications: Data Science with python (Devspark in collaboration with Microsoft), **Data Science hands-on capstone (IBM)**, python certification(HackerRank, Cisco), AWS certificate(EduSkills), Data Structures Silver Badge(VCET), Tableau Business Intelligence certificate(Coursera – In progress).
Contributions: Co-authored a research paper on “**Deep Neural Network Ensembles for the Detection of Alzheimer’s Disease using Imbalanced Clock Drawing Test Images**” in Computational Intelligence at **IEEE** Active Contributor on Hacker Rank, Quora Answerer, Q&A Expert on Chegg, Volunteered IGSA.

Work Experience

Data Scientist | University of Michigan – Dearborn, MI, USA May 2023 – Present

- Conducted pioneering research with the NIH (National Institute of Health), studying driving behavior signatures to classify Alzheimer's onset across various age groups.
- Developed an innovative automated tool to annotate driving categories and analyze vehicle speeds, reducing 90% of manual work.
- Operated an automated system to efficiently sort daily recorded signals for each participant, streamlining data management processes.
- Analyzed clock drawing data from NHATS using Stata, developing code to decode and interpret behavioral patterns for advanced data-driven insights.
- Created cutting-edge deep learning algorithms using ResNet, ViT, and EfficientNet to classify clocks based on Alzheimer’s severity, achieving 85-89% accuracy despite data bias. Utilized Bootstrap and oversampling techniques to handle unbalanced data.
- Assisted in conducting a driving study with Nissan, achieving 87% accuracy in estimating vehicle speeds. Currently preparing a publication on the findings.
- Developed complex SQL scripts for querying and preprocessing large vehicle dataset. Utilized Microsoft Excel to visualize data and perform intricate calculations.

Tata Consultancy Services – Hyderabad, India. Assistant Engineer Jun 2021 – Aug 2022

- Utilized Python scripts to automate monitoring of DevOps tools, significantly enhancing system efficiency, reducing operational overhead, and integrating CI/CD pipeline notifications for streamlined operations.
- Promptly diagnosed and resolved DevOps tools-related issues, ensuring minimal disruption to CI/CD workflows, and maintaining optimal system performance.
- Designed and implemented comprehensive disaster recovery strategies, including rigorous testing and validation, to ensure business continuity during system failures.
- Efficiently managed platform operations on Jira and Confluence, maintaining a smooth work environment. Automated data extraction and loading (ETL) processes to improve data management and accessibility, facilitating data-driven decision-making.
- Utilized Excel for advanced data analysis, including use of pivot tables, complex formulas and VBA scripts to automate tasks.
- Leveraged advanced data integration techniques to enhance the accessibility and usability of data across platforms, supporting critical data analysis and machine learning initiatives.

Internshala – Remote. Intern Jul 2020 – Sep 2020

- Engineered a sophisticated Fantasy Cricket application using Python, predicting winning probabilities based on real-time player performance data.
- Leveraged sqlite3 to integrate a robust database into the application, optimizing memory usage by 70% for enhanced performance and efficiency
- Implemented the PyQt5 library to design an intuitive and user-friendly GUI, seamlessly integrating front-end technologies like HTML and CSS within Python.

ECIL – Hyderabad. Intern Apr 2019 – Jul 2019

- Engineered and optimized SQL queries for data extraction and preprocessing, while developing dynamic Power BI dashboards and reports that integrated various data sources. Enhanced data models using advanced DAX functions to facilitate complex analytical calculations.
- Performed rigorous data validation and cleansing using SQL and Excel to maintain consistency across reports, collaborating with cross-functional teams to establish data requirements and implementing tailored Power BI solutions to provide actionable business insights.

Academic Projects

Road Lane-Curve Detection System: (Language used: Python) Apr 2023

- Developed a CNN-based model using cloud services for lane detection, lane-keeping, and road curvature analysis.
- Utilized cloud services like the Video Intelligence API to process live streaming video, minimizing memory usage.
- Achieved 92.5% accuracy in displaying lane-keeping status and road curvature radius in real-time.

Prediction System of Crops with Machine learning: (Language used: python) Dec 2022

- Developed a crop selection system to recommend the best crops for specific soil and climatic conditions.
- Implemented and compared six models: KNN, Decision Trees, Random Forest Naïve Bayes, XGBoost and SVM with accuracy of 95%.

Sign Language Recognition using Convolution Neural Network: (Language used: Python) Jun 2021

- Created a CNN-based model to translate sign language into text, facilitating communication for deaf mute individuals.
- Achieved 98.7% accuracy in converting sign language to text. This model is published in Springer Publications, accepted in 2021.

Web Scrapping: (Language used: Python) Jan 2021

- Developed a project to preprocess and convert unstructured web data into structured format.
- The model efficiently extracts valuable content from webpages with an accuracy of 86%.

Education

University of Michigan Aug 2022 - Apr 2024

Masters in data science

Lakireddy Bali Reddy College of Engineering, India Jun 2017 - Jun 2021

Bachelor of Technology in Computer Science Engineering